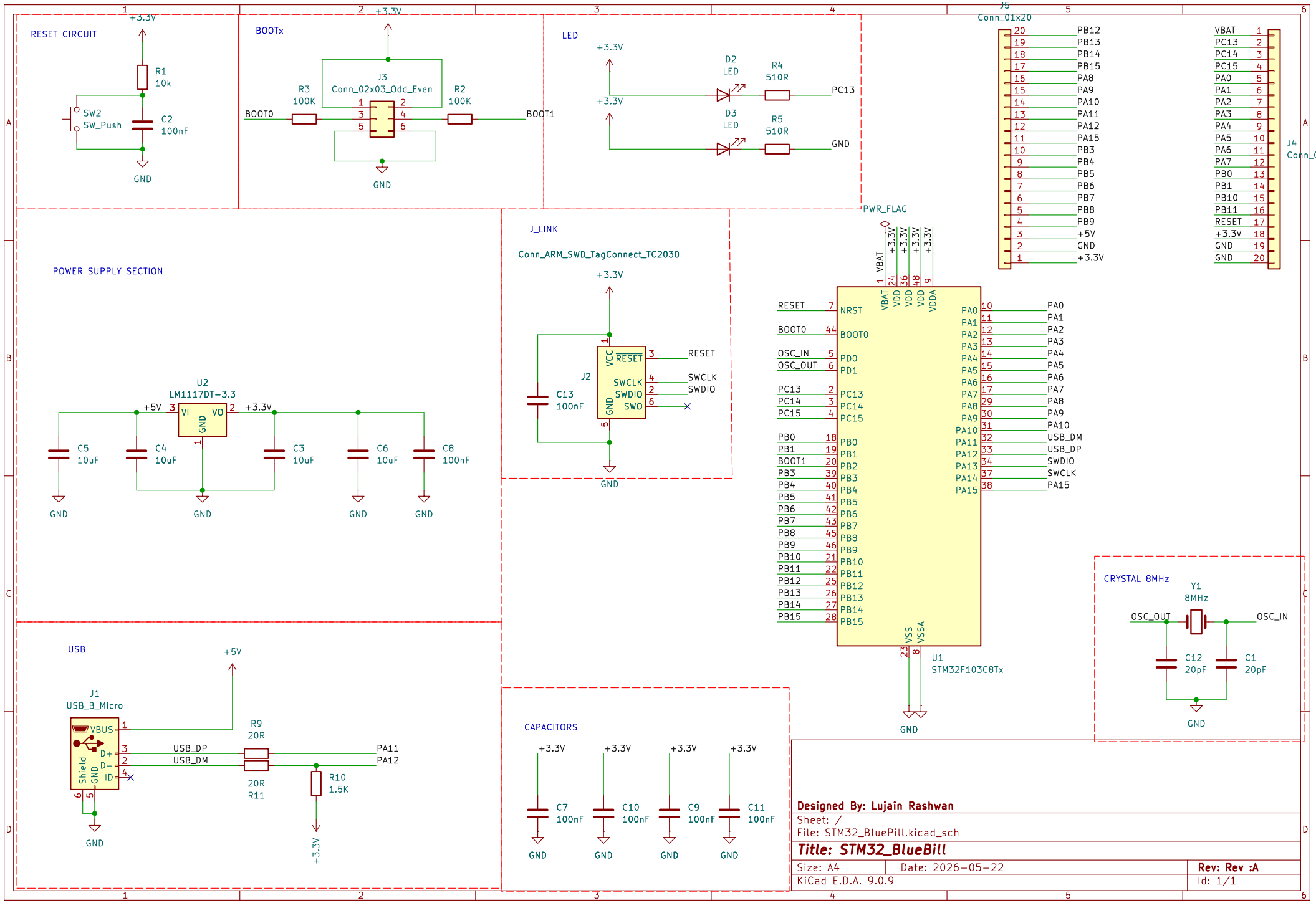
RESET CIRCUIT
+3.3V
R1
10k
SW2
SW_Push
C2
100nF
GND
BOOTx
+3.3V
J3
Conn_02x03_Odd_Even
R3
100K
R2
100K
BOOT0
BOOT1
GND
LED
+3.3V
D2
LED
R4
510R
PC13
+3.3V
D3
LED
R5
510R
GND
J5
Conn_01x20
20 PB12
19 PB13
18 PB14
17 PB15
16 PA8
15 PA9
14 PA10
13 PA11
12 PA12
11 PA15
10 PB3
9 PB4
8 PB5
7 PB6
6 PB7
5 PB8
4 PB9
3 +5V
2 GND
1 +3.3V
J4
Conn_01x20
VBAT 1
PC13 2
PC14 3
PC15 4
PA0 5
PA1 6
PA2 7
PA3 8
PA4 9
PA5 10
PA6 11
PA7 12
PB0 13
PB1 14
PB10 15
PB11 16
RESET 17
+3.3V 18
GND 19
GND 20
POWER SUPPLY SECTION
U2
LM1117DT-3.3
+5V
VI
VO
GND
+3.3V
C5
10uF
C4
10uF
C3
10uF
C6
10uF
C8
100nF
GND
J_LINK
Conn_ARM_SWD_TagConnect_TC2030
+3.3V
J2
VCC
RESET
SWCLK
SWDIO
SWO
GND
C13
100nF
RESET
SWCLK
SWDIO
GND
PWR_FLAG
VBAT
+3.3V
VDD
VDD
VDD
VDDA
RESET 7 NRST
BOOT0 44 BOOT0
OSC_IN 5 PD0
OSC_OUT 6 PD1
PC13 2 PC13
PC14 3 PC14
PC15 4 PC15
PB0 18 PB0
PB1 19 PB1
BOOT1 20 PB2
PB3 39 PB3
PB4 40 PB4
PB5 41 PB5
PB6 42 PB6
PB7 43 PB7
PB8 45 PB8
PB9 46 PB9
PB10 21 PB10
PB11 22 PB11
PB12 25 PB12
PB13 26 PB13
PB14 27 PB14
PB15 28 PB15
PA0 10 PA0
PA1 11 PA1
PA2 12 PA2
PA3 13 PA3
PA4 14 PA4
PA5 15 PA5
PA6 16 PA6
PA7 17 PA7
PA8 29 PA8
PA9 30 PA9
PA10 31 PA10
PA11 32 USB_DM
PA12 33 USB_DP
PA13 34 SWDIO
PA14 37 SWCLK
PA15 38 PA15
VSS
VSSA
23
8
U1
STM32F103C8Tx
GND
CRYSTAL 8MHz
Y1
8MHz
OSC_OUT
OSC_IN
C12
20pF
C1
20pF
GND
USB
+5V
J1
USB_B_Micro
VBUS
D+
D-
ID
GND
Shield
USB_DP
USB_DM
R9
20R
20R
R11
R10
1.5K
PA11
PA12
+3.3V
CAPACITORS
+3.3V
C7
100nF
C10
100nF
C9
100nF
C11
100nF
GND
Designed By: Lujain Rashwan
Sheet: /
File: STM32_BluePill.kicad_sch
Title: STM32_BlueBill
Size: A4
Date: 2026-05-22
Rev: Rev :A
KiCad E.D.A. 9.0.9
Id: 1/1